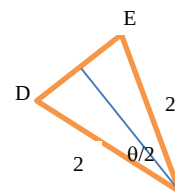


Section A: Pure Mathematics (40 marks)

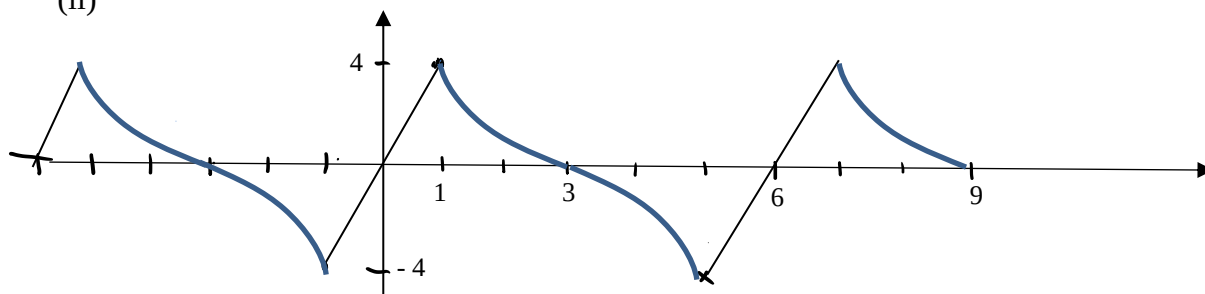
1 $DE = 2 \left[2 \sin \frac{\theta}{2} \right] = 4 \sin \frac{\theta}{2}$ Alternative, use cosine rule

Perimeter $= 4 \times 2 + 2 \left[4 \sin \frac{\theta}{2} \right] = 8 \left[1 + \sin \frac{\theta}{2} \right]$

Unit cost $= \frac{28000}{8 \left[1 + \sin \frac{\theta}{2} \right]} = 3500 \left(1 + \sin \frac{\theta}{2} \right)^{-1}$
 $\approx 3500 \left(1 + \frac{\theta}{2} \right)^{-1}$
 $= 3500 \left(1 - \frac{\theta}{2} \dots \right)$
 $\approx 3500 - 1750\theta$



2 (i) $f(4) = f(-2) = -f(2) = -1$.
(ii)



(iii) $\int_{-5}^7 |f(x)| dx = 4 \left[\frac{1}{2} \times 4 \times 1 + \int_1^3 (3-x)^2 dx \right]$
 $= 4 \left[2 + \left[\frac{(3-x)^3}{-3} \right]_1^3 \right]$
 $= 8 - \frac{4}{3} [0 - 8] = \frac{56}{3}$

3 line $l : 2-x = \frac{y-3}{3}, z=2 \Rightarrow \vec{r} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix} \quad \lambda \in \mathbb{R}$

(i) $\overrightarrow{CD} = \begin{pmatrix} 3 \\ \alpha \\ \beta \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ \alpha-3 \\ \beta-2 \end{pmatrix} = k \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix}$

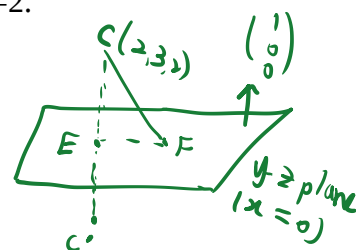
Hence $k = -1$, $\alpha = 0$ and $\beta = 2$

(ii) Equation of y - z plane is $x = 0$.

At the point of intersection F of the line l and the y - z plane, $2-\lambda=0 \Rightarrow \lambda=2$.

Hence $F(0, 9, 2)$

(iii) By observation, $\overrightarrow{OE} = \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix}$ and hence $\overrightarrow{OC} = \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix}$



Alternative method: Let E be the foot of $\perp$ from C to y-z plane $\vec{r} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$

$$\text{Line CE: } \vec{r} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2+\mu \\ 3 \\ 2 \end{pmatrix}.$$

$$\text{When line CE meets y-z plane: } \vec{r} = \begin{pmatrix} 2+\mu \\ 3 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0 \Rightarrow \mu = -2. \therefore \vec{OE} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + (-2) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix}$$

$$\text{By mid-point theorem, } \vec{OE} = \frac{\vec{OC} + \vec{OC'}}{2} \Rightarrow \vec{OC'} = 2\vec{OE} - \vec{OC} = 2 \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix}$$

4 (a) $\frac{dy}{dx} + 4(x-y)^2 \cos^2 x = \sin^2 x$

$$\text{Using } u = x - y, \quad \frac{du}{dx} = 1 - \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = 1 - \frac{du}{dx}$$

$$1 - \frac{du}{dx} + 4u^2 \cos^2 x = \sin^2 x$$

$$\frac{du}{dx} = 4u^2 \cos^2 x + 1 - \sin^2 x$$

$$\frac{du}{dx} = (1 + 4u^2) \cos^2 x$$

$$\int \frac{1}{1 + 4u^2} du = \frac{1}{2} \int (1 + \cos 2x) dx$$

$$\frac{1}{4} \int \frac{1}{\frac{1}{4} + u^2} du = \frac{1}{2} \int (1 + \cos 2x) dx$$

$$\frac{1}{4} [2 \tan^{-1} 2u] = \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) + c$$

$$\tan^{-1} [2(x-y)] = x + \frac{1}{2} \sin 2x + c$$

(b)(i) $\frac{dx}{dt} = R - kx^2$, k is a positive constant

$$\text{At } x = 2R, \quad \frac{dx}{dt} = 0$$

$$0 = R - k(2R)^2 \Rightarrow 0 = R(1 - 4kR) \Rightarrow k = \frac{1}{4R}$$

$$\frac{dx}{dt} = R - \frac{x^2}{4R}$$

$$\int \frac{4R}{4R^2 - x^2} dx = \int 1 dt$$

$$\ln \left| \frac{2R+x}{2R-x} \right| = t + C$$

$$\frac{2R+x}{2R-x} = \pm e^C e^t \Rightarrow \frac{2R+x}{2R-x} = A e^t, A = \pm e^C$$

At $t=0, x=0 \Rightarrow A=1$ Hence $\frac{2R+x}{2R-x} = e^t \Rightarrow x = 2R \left[\frac{e^t - 1}{1 + e^t} \right]$

(b)(ii) $x = 2R \left[\frac{1 - \frac{1}{e^t}}{\frac{1}{e^t} + 1} \right]$ As $t \rightarrow \infty, \frac{1}{e^t} \rightarrow 0 \Rightarrow x \rightarrow 2R$ [or use graphical method]

The amount of drug in the body will not exceed $2R$ mg regardless of the period of treatment.

5 $z^3 = \frac{1}{\sqrt{2}} \left(1 + \frac{2}{2a-1}i \right)$

$\arg(z^3) = -\frac{\pi}{4} \Rightarrow \frac{2}{2a-1} = -1 \Rightarrow a = -\frac{1}{2}$

(i) $z^3 = \frac{1}{\sqrt{2}}(1-i) \Rightarrow z^3 = e^{i\left(-\frac{\pi}{4} + 2k\pi\right)}$, $k = 0, +1, -1$

$z = e^{i\left(-\frac{\pi}{12} + \frac{2k\pi}{3}\right)}$, $k = 0, \pm 1$

$z = \cos\left(\frac{-\pi}{12}\right) + i \sin\left(\frac{-\pi}{12}\right)$, $z = \cos\left(\frac{-3\pi}{4}\right) + i \sin\left(\frac{-3\pi}{4}\right)$ and $z = \cos\left(\frac{7\pi}{12}\right) + i \sin\left(\frac{7\pi}{12}\right)$

$\sqrt{2} \left(\frac{w}{1+i} \right)^3 = 1-i \Rightarrow \left(\frac{w}{\sqrt{2}e^{i\frac{\pi}{4}}} \right)^3 = \frac{1-i}{\sqrt{2}}$

$\frac{w}{\sqrt{2}e^{i\frac{\pi}{4}}} = z \Rightarrow w = \sqrt{2}e^{i\frac{\pi}{4}}z$

The points representing w is an anti-clockwise rotation of the points representing z about the origin by $\frac{\pi}{4}$, followed by an enlargement by factor $\sqrt{2}$ about the origin.

Section B: Statistics (60 marks)

6 PP EEE C R T I V

i) number of ways = $\frac{6! \times 7 \times 6 \times 5 \times 4}{3! \times 2!} = 50400$

ii) P _ _ P _ _ _ _ _

Case 1: 2 E $7! = 5040$

Case 2: 1 E $\left({}^5C_1 \times 2!\right) \times \frac{7!}{2!} = 25200$ Total = 30240

7. (a) (i) $P(A \cup B) \leq 1$ and $P(A \cap B) = P(A) + P(B) - P(A \cup B)$
 $\geq 0.7 + 0.8 - 1$
 $= 0.5$

(i) $P(A' \cup B) + P(A) - P(A \cap B) = 1$ (Use Venn Diagram)

$P(A \cap B) = 0.85 + 0.7 - 1 = 0.55$

$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.55}{0.8} = \frac{11}{16} = 0.6875$

A & B are not independent as $P(A|B) \neq P(A)$. or $P(A \cap B) \neq P(A).P(B)$

(ii) If $P(A \cap B) = 0.5$, then $P(A \cup B) = 1$, event A & event B are exhaustive.

(b) Total no. of points = $1+2+3+\dots+9+10=55$ OR ${}^{11}C_2 = 55$

$$\text{Prob.} = {}^{11}C_1 \times \frac{10}{55} \times \frac{9}{54} \times \frac{8}{53} = \frac{8}{159} \approx 0.0503 \quad \text{OR} \quad {}^{11}C_1 \times \frac{{}^{10}C_3}{{}^{55}C_3} = \frac{8}{159}$$

8 i) Let X = number of hot drinks sold in five minutes.

Let Y = number of cold drinks sold in five minutes

$$X \sim Po(2) \quad Y \sim Po(2.5)$$

$$X+Y \sim Po(4.5)$$

$$\begin{aligned} P(X+Y > 4) &= 1 - P(X+Y \leq 4) \\ &= 1 - 0.532103\dots \\ &= 0.467896\dots = 0.468 \end{aligned}$$

ii) Required probability = $P(X=0 / X+Y \leq 4)$

$$\begin{aligned} &= \frac{P(X=0 \cap X+Y \leq 4)}{P(X+Y \leq 4)} \\ &= \frac{P(X=0)P(Y \leq 4)}{P(X+Y \leq 4)} = \frac{0.13534 \times 0.89118}{0.53210} = 0.227 \end{aligned}$$

(iii) Let W = number of periods with more than 4 drinks sold out of the twenty periods.

$$W \sim B(20, 0.46790)$$

$$P(W \geq n) < 0.4$$

$$1 - P(W \leq n-1) < 0.4 \Rightarrow P(W \leq n-1) > 0.6$$

$$\text{From GC: } P(W \leq 9) = 0.527 < 0.6$$

$$P(W \leq 10) = 0.696 > 0.6$$

$$\text{Least } n-1 = 10 \quad \Rightarrow \text{least } n = 11$$

(iv) Number of hot drinks sold in 30 minutes = $H \sim Po(12)$

Number of cold drinks sold in 30 minutes = $C \sim Po(15)$

Since both $\lambda > 10$, $H \sim N(12, 12)$ and $C \sim N(15, 15)$

$$H - C \sim N(-3, 27)$$

$$P(H - C > 0) = P(H - C > 0.5) = 0.25029\dots = 0.250$$

$$9(a) \quad n=16 \quad \bar{x} = \frac{2848}{16} = 178,$$

$$s^2 = \frac{1}{15} \left[509884 - \frac{2848^2}{16} \right] = 196, \quad s = 14$$

Hypothesis: $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$

As σ^2 is unknown, n is small, conduct a 2-tailed t -test,

assuming $X \sim \text{Normal Distribution } N(\mu, \sigma^2)$, test statistics: $T = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{16}}} \sim t(15)$

At 5% significance level, do not reject H_0 if $-2.131 < t < 2.131$.

$$-2.131 < \frac{178 - \mu_0}{\frac{14}{\sqrt{16}}} < 2.131$$

$$-7.4585 < 178 - \mu_0 < 7.4585$$

$$170.5415 < \mu_0 < 185.4585 \Rightarrow 171 < \mu_0 < 185$$

(b) Combine sample: $n = 16 + 36 = 52$, $120 = \frac{1}{36} \left[\sum c^2 - \frac{6660^2}{36} \right] \Rightarrow \sum c^2 = 1236420$

$$\sum y = 2848 + 185(36) = 9508$$

and $\sum y^2 = 509884 + 1236420 = 1746304$.

Unbiased estimate for μ is $= \bar{y} = \frac{9508}{52} = 182.8461538$

Unbiased estimate for σ^2 is $= s^2 = \frac{1}{51} \left[1746304 - \frac{9508^2}{52} \right] = 152.9954751$

Hypothesis: $H_0 : \mu = 180$ against $H_1 : \mu > 180$

As σ^2 is unknown, n is large, conduct a 1-tailed z -test,

test statistics: $Z = \frac{\bar{X} - 180}{\frac{s}{\sqrt{52}}} \sim N(0,1)$ [by CLT]

From GC, $z = 1.65928407$, $p = 0.04852928$

p -value $= P(Z > 1.65928407)$ when $\mu = 180$

= prob that the sample mean cholesterol attains a value of more than 182.8461538 if the population mean cholesterol content in eggs is 180

Or The p -value is the lowest level of significance for which the null hypothesis of mean cholesterol level being 180, will be rejected.

Or The p -value is the probability of obtaining a test statistic more than 1.659, assuming that H_0 is true.

10 Let L = waiting time of a patient at Lee's Clinic and H = waiting time of a patient at Hope Clinic.

$$L \sim N(25, 8^2) \quad H \sim N(37, 4^2)$$

i) Let $X = \frac{L_1 + L_2 + L_3 + L_4 + L_5}{5} \sim N\left(25, \frac{8^2}{5}\right)$

$$X - H \sim N\left(25 - 37, \frac{8^2}{5} + 4^2\right) = N(-12, 28.8)$$

Required probability $= P(|X - H| \geq 5)$

$$= P(X - H \geq 5) + P(X - H \leq -5)$$

$$= 0.000768 + 0.90395$$

$$= 0.905 \text{ (to 3 sf)}$$

Assumption: the waiting times of all patients are independent.

ii) Let Y = number of patients with waiting time more than 25 minutes

$$Y \sim B(n, 0.5)$$

Since $n > 40$, $\therefore np > 5$ and $n(1-p) > 5$

$\Rightarrow Y \sim N(0.5n, 0.25n)$ approximately.

$$P(Y \leq 40) < 0.95$$

$$\Rightarrow P(Y \leq 40.5) < 0.95$$

$$\Rightarrow P\left(Z \leq \frac{40.5 - 0.5n}{0.5\sqrt{n}}\right) < 0.95$$

$$\Rightarrow \frac{40.5 - 0.5n}{0.5\sqrt{n}} < 1.64485$$

$$\Rightarrow 0.5n + 1.64485(0.5\sqrt{n}) - 40.5 > 0$$

By GC, $n > 67.487$. Least $n = 68$

iii) Let W = number of patients treated for influenza in a sample of 20.

$$W \sim B(20, 0.2)$$

$$E(W) = 20(0.2) = 4 \quad \text{and} \quad \text{Var}(W) = 20(0.2)(0.8) = 3.2$$

Since $n = 60$, by CLT, $\bar{W} \sim N\left(4, \frac{3.2}{60}\right)$

$$P(\bar{W} > 3.5) = 0.9848... = 0.985$$

iv) Normal model is not likely to be appropriate as the waiting times of the combined group of patients may follow a bi-modal distribution.

11. (i) $\bar{\theta} = 45$, $\bar{T} = \frac{80+t}{7}$ lies on the regression line.

$$\frac{80+t}{7} = 35.857 - 0.51429(45) \Rightarrow t = 8.998 \approx 9 \text{ (nearest integer)}$$

(ii) When $\theta = 70$, $T = -0.143$.

Reason 1 : this estimate is not realistic since the time taken cannot be negative.

Reason 2: scatter diagram suggests a curved relationship rather than a straight line.

$$(iii) \theta(T - a) = b \Rightarrow T = a + \frac{b}{\theta}$$

$$\text{Regression line of } T \text{ on } \frac{1}{\theta}: T = -8.1556 + \frac{804.86}{\theta}$$

$$30 = -8.1556 + \frac{804.86}{\theta} \Rightarrow \theta = 21.1^\circ\text{C}$$

[Note: the wrong regression line gives 21.2°C]